

PPT Content

Great — I've reviewed your PPT structure from the uploaded file .

Since you've upgraded from **PHQ-9 questionnaire** → **structured daily mental health metrics logging + AI dashboard**, your PPT content must reflect:

- Shift from diagnostic questionnaire → continuous behavioral analytics
- Data-driven insights instead of static scoring
- AI + local LLM therapist integration
- Quantifiable metrics dashboard

Below is the **updated slide-wise content** formatted exactly according to your existing PPT sections.

You can copy-paste slide by slide.

● TITLE SLIDE

(Keep as is — no major changes needed)

MentraAI

An Intelligent System for Mental Health Assessment and Guidance

PBL Theme: HealthCare

Department of Computer Engineering

● ABSTRACT (UPDATED)

MentraAI is an AI-powered mental health monitoring system that replaces traditional questionnaire-based assessments with structured daily mental health metric logging. Instead of relying on static forms such as PHQ-9, the system collects quantifiable daily indicators including mood, stress, anxiety, energy, sleep quality, social connection, and productivity.

The collected data is analyzed using statistical methods and AI models to generate personalized insights, detect behavioral patterns, and provide proactive recommendations. The platform also integrates a locally hosted Large Language Model (LLM) therapist for conversational support while maintaining privacy. MentraAI aims to make mental wellness tracking continuous, data-driven, and accessible.

● INTRODUCTION (UPDATED)

Mental health assessment traditionally relies on standardized questionnaires such as PHQ-9 and GAD-7. While effective in clinical settings, these tools are periodic and static.

MentraAI introduces a continuous mental health monitoring approach by:

- Logging structured daily mental health metrics
- Visualizing trends through dashboards
- Computing derived insights such as burnout and isolation risk

- Providing AI-powered conversational therapy
- Ensuring privacy through local LLM deployment

This system shifts from episodic assessment to continuous self-reflection and pattern detection.

● LITERATURE SURVEY / RESEARCH

1. Traditional Assessment Models
 - PHQ-9, GAD-7
 - Episodic and questionnaire-based
 2. Behavioral Data Tracking
 - Continuous mood and sleep tracking improves mental health awareness
 3. AI in Mental Health
 - LLM-based conversational therapy
 - Sentiment analysis for emotional understanding
 4. Digital Mental Health Systems
 - Dashboard-based visualization improves user engagement
 - Data-driven mental health predictions
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● PROPOSED WORK (UPDATED TO DAILY METRICS MODEL)

MentraAI proposes a structured daily logging system that captures:

- Mood (1–10)
- Anxiety Level (1–10)
- Stress Level (1–10)
- Energy Level (1–10)
- Sleep Hours
- Sleep Quality (1–10)
- Social Connection (1–10)
- Productivity (1–10)

The system processes this data to compute:

❶ Emotional Stability Index
(Standard deviation of mood over time)

❷ Burnout Risk
(High stress + low energy + poor sleep)

❸ Isolation Risk
(High anxiety + low social connection)

Additionally, a locally deployed GGUF LLM via llama-server provides therapeutic conversational support.

● SYSTEM ARCHITECTURE (Replace UML Slide Content)

Frontend:

- HTML, CSS, Vanilla JavaScript
- Journal logging interface
- Dashboard visualization (Chart.js)

Backend:

- Python (FastAPI)
- REST API endpoints
- SQLite database

AI Layer:

- GGUF model
- llama-server for local inference
- AI therapist prompt engineering

Data Flow:

User Input → Backend API → Database + AI Processing → Dashboard Insights

● WORK CARRIED OUT TILL DATE (TABLE FORMAT)

You can paste this in table format:

Sr	Work Description	Status
1	Designed UI for Journal & Dashboard	Completed
2	Implemented Daily Metric Logging	Completed
3	Integrated Local LLM Therapist	Completed
4	Developed Dashboard Visualizations	Completed
5	Implemented Derived Metric Calculations	Completed
6	Connected Frontend with Backend APIs	Completed
7	Database Schema Design	Completed

● UI DESIGNS

Replace PHQ-9 reference with:

- Daily structured logging interface
- Slider-based emotional inputs
- Dashboard cards and analytics
- Trend graphs (Mood, Stress vs Sleep, etc.)
- Derived insight panel

● FUTURE WORK (UPDATED TO MATCH NEW MODEL)

Multimodal Input Integration

- Voice-based journaling

- Speech emotion detection

Advanced AI Analytics

- Time-series prediction models
- Correlation coefficient analysis
- Personalized mental health scoring

Predictive Modeling

- Early burnout detection
- Anxiety spike forecasting

Cloud & Security

- Encrypted multi-device synchronization

Wearable Integration

- Heart rate variability
- Sleep cycle tracking
- Activity data

● CONCLUSION (UPDATED)

MentraAI demonstrates a transition from static questionnaire-based mental health assessment to continuous data-driven behavioral analytics.

By combining:

- Structured daily metric logging
- Statistical trend analysis
- Derived mental wellness indicators
- AI-powered conversational therapy
- Privacy-focused local LLM deployment

The system empowers individuals to monitor, understand, and proactively manage their mental health.

MentraAI bridges the gap between self-reflection tools and intelligent AI assistance.

● REFERENCES (Keep Most Same + Minor Adjustment)

You may keep your existing references but emphasize:

- LLaMA / GGUF architecture
- Digital Mental Health Informatics
- Time-series behavioral analysis

● THANK YOU SLIDE

No change required.

□ What Changed Conceptually

Before:

Static PHQ-9 questionnaire scoring

Now:

Continuous mental health analytics system with AI-assisted insights

This sounds MUCH stronger academically.

If you want, I can next:

- Help you rewrite it in more formal academic tone
- Create short speaker notes for viva
- Help you defend the shift from PHQ-9 in presentation
- Generate architecture diagram explanation text
- Help you prepare technical questions & answers

What do you want to prepare for next — viva or final demo?